

FORM FOUR EXAMS SET 3



*FOR ANSWERS AND MORE SETS OF EXAMS, SMS OR
WHATSAPP: 0723981225*

AGRICULTURE

FORM 4

TERM 1, 2020

Answer all questions

1. Define egg candling as used in poultry production (1mk)
2. State four conditions necessary for artificial incubation (2mks)
3. Highlight any five advantages of battery cage system in poultry production (5mks)
4. State any two causes of stress in poultry and for each give a control measures (2mks)
5. State the different ways of controlling cannibalism among layers in deep litter system (4mks)
6. What qualities should eggs meant for incubation meet in poultry production? (3mks)

7. a) What is colostrum ($\frac{1}{2}$ mk)

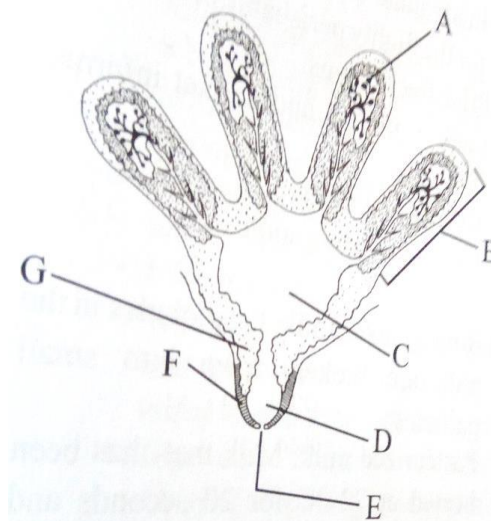
b) Give four qualities of colostrum that make it suitable for new born calves (2mks)

c) State the components of artificial colostrum (2mks)

8. State the characteristics of clean and high quality milk(2mks)

9. State four factors that determine the quality of hay (2mks)

10. The diagram below shows the structure of the udder, study it and answer the questions that follow



a) State the parts labeled A, B, C, D, E, F (3mks)

b) In which of the labeled parts does secretion take place (1mk)

c) Name the two hormones that control milk secretion (1mk)

d) For how long should milking process take? (1mk)

e) Give a reason for your answer in (d) above (1mk)

11. State two precautions observed in harvesting of tea (2mks)

12. State two ways of weed control in rice production (1mk)

13. Name three methods of conserving silage(1½mks)

14. State any four bacterial diseases affecting livestock (2mks)

15. Explain five requirements of a good calf pen (5mks)

16. a) Define layering in crop production (1mk)

c) Name four methods of layering (2mks)

17. A dairy farmer wanted to prepare 1000kg of a calf ration containing 20% DCP. The feedstuffs available were maize (10% DCP) and Sunflower (35% DCP). Using Pearson Square method, calculate the amount of each of the rearing ration (3mks)

BUSINESS STUDIES

TERM 1 YEAR 2020

FORM FOUR

NAME.....ADM.....CLASS.....

1. Highlight five importance of sources documents in business.(5mks)

2. For each of the following transactions indicate the books of original entry which they would be recorded.(5mks)
 - a) Bought goods on credit.

 - b) Sold an old typewriter on credit.

 - c) Returned goods bought on credit for sale.

 - d) Opening and closing entries.

 - e) When cash is received for goods sold.

- 3.a) Explain five types of errors that may not be detected by a trial balance.(10mks)

b) State five features of an efficient transport system.(5mks)

4. a) State four characteristics of basic human wants. (4mks)

b) State five reasons why most businesses in Kenya operate on small scale. (5MKS)

5. a) In each of the following cases state the sources documents used.(4mks)

- i)When goods are sold in cash.
- ii)When correcting an under-charge.
- iii) When goods are returned by a customer
- iv) When the petty cashier pays for expenses in an office

b) The following transactions took place in the Mali Traders in the month of June 2016.

June 4: Sold goods on credit to Biyang sh14,000 and Wendy sh17000

June 7; Bought goods on credit from Faddy sh43,000

June 10; Biyang returned goods worth sh3000

June12; Purchased goods on credit from Violet sh64,000 and Elastia sh28,000.

June 18; Goods worth sh5000 were returned to Faddy and goods worth sh2000 were returned to Violet.

June 22; Goods sold on credit to Biyangsh 6000,Amwangush 10,000 and Josephine sh 11,000

June 28; Goods returned to Amwangush 10,000.

Required: Prepare the relevant Journals. (12mks)

NAMEADM NO.....CLASS.....

FORM 4 CRE

TERM 1 2020

1. State the sources of Christian ethics (5mks)
2. Explain the traditional African understanding of human sexuality(7mks)
3. State the consequences of irresponsible sexual behavior(6mks)
4. Explain why Christians practice celibacy(6mks)
5. State 5 importance of courtship (5mks)
6. Explain 6 reasons why people should work (6mks)
7. State 5 moral duties and responsibilities of employers (5mks)
8. Outline Nathans prophecy to king David about the messiah(5mks)
9. Identify 5 factors contributing to harmony and mutual responsibility in African Traditional society (5mks)

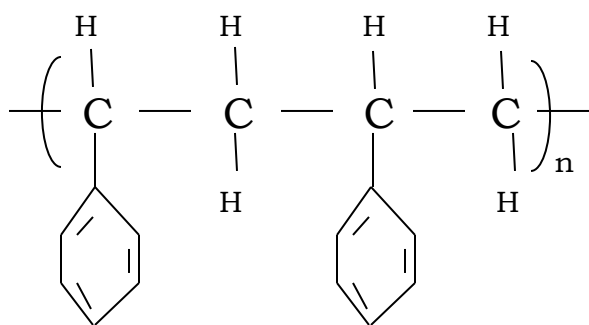
NAME:CLASS: ADM.NO. :

CHEMISTRY FORM FOUR

TERM 1 -2020

1. (a) Distinguish between a strong and a weak acid. Give example to each. (2mks)

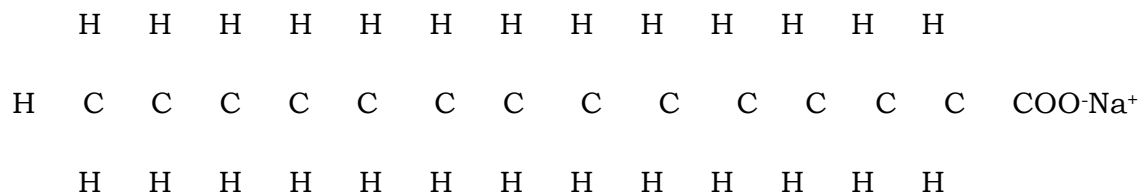
(b) The formula given below represents a portion of a polymer. Give ;



i) The name of the polymer. (1mk)

ii) Two disadvantage of continued use of this polymer. (2mks)

2. (a) The structure of a detergent is.



i) Write the molecular formula of the detergent. (1mk)

ii) What type of detergent is represented by the formula? (1mk)

iii) When this type of detergent is used to wash linen in hard water, spots (marks) are left on the linen. Write the formula of the substance responsible for the spot. (1mk)

(b) Give the systematic names of the following compound.

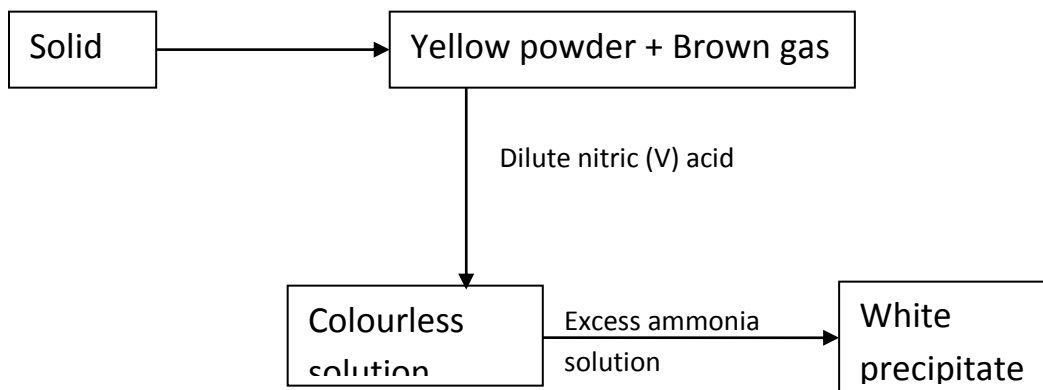
i) $\text{CH}_3\text{CH}_2\text{COOH}$. (1mk)

ii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_2$. (1mk)

iii) $\text{CHCCH}_2\text{CH}_3$. (1mk)

iv) $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}_3$. (1mk)

3. (a) Study the flow chart below and answer the questions that follow.



a. Give the chemical formula of
i) Solid P (1mk)

ii) The yellow powder. (1mk)

iii) The white precipitate. (1mk)

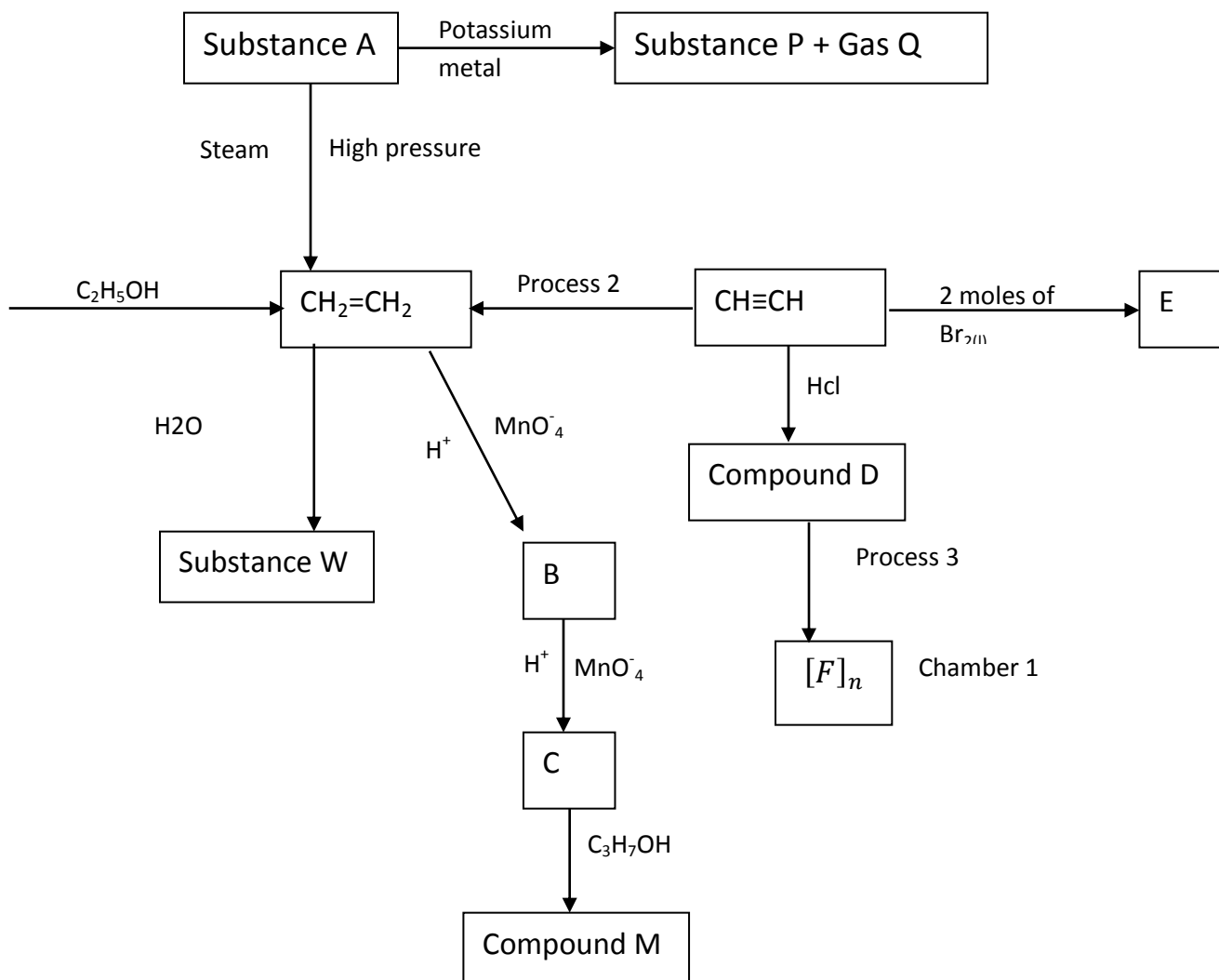
b. (i) Identify ions that cause water hardness.

(2mks)

(ii) List the two types of water hardness.

(2mks)

c. Study the flow chart below and use it to answer the questions that follow.



i. Identify substances:-

(6mks)

A –

W –

P –

E –

D –

Gas Q –

- ii. State the conditions and reagents that are required for processes. (6mks)
Process 1 –

Process 2 –

Process 3 –

- iii. Write the formula and the name of the compound formed in chamber I. (2mks)

- iv. Give the uses of the compound in c(iii) above. (2mks)

- d. State the physical property of compound M and its structural formula. (3mks)

4. (a) (i) The reaction of calcium with sulphuric (VI) acid starts and then comes to stop. Explain. (3mks)

- (ii) Starting with copper(II) oxide, describe how a solid sample of copper (II) sulphate filtrate would be prepared. (4mks)

(b) (i) Define solubility. (1mk)

(ii) 20 grams of potassium chlorate saturated 20cm³ of water. Calculate its solubility.
(3mks)

(iii) Define the term fractional crystallization of salts. (1mk)

NAME.....ADM NO.....CLASS...

ENGLISH FORM 4

TERM 1 -2020

TIME 1hr

Fill in each blank space in the passage below with the most appropriate word. (10mks)

Good health is a key a component to a fulfilled ____1____. Most definitions of health ____2____ not only the individual's physical and mental well-being , but also healthy ____3____ with the surrounding society and environment .It is ____4____ critically important to personal good health that ____5____ and environmental conditions be supportive ____6____ individual physical and mental well-being .The health of the ____7____ and that of the individual are ____8____ linked. Health-care workers typically ____9____ on individual aspects of the patient's physical or mental health, ____10____ remain very cognizant of the effects that the home , family , or social economic status have on the outcome of the care prescription.

2.Read the poem below and answer the questions that follow.(20mks)

You embarrass me

Mwananchi

Why do you embarrass me with your questions

About the new Mercedes I bought

The large farm I own

The houses , the wives

An inflated stomach !

Mwananchi

Why do you frighten me with your threats

The threats in your bloodshot eyes

Fixedly pointed at me wherever I go

Like if you are ready

To release the arrow that will deflate me

Into nothingness;

Even the watchmen , the dogs , the police
Are all not enough to protect me
From your increasing shouts to protest
Against my good judgement ;

Mwananchi

Have you forgotten how you loved me
And gave me your vote
That I may be your man in parliament?

Now that I have the power
I will mend your confused senses
And keep you in prison
Until you see me as your leader again
And keep those bloodshot eyes away from me

I will charge like an angry lion And scare you out of your wits
Until like a frightened dog ,
You keep your head down forever ...

Everett M .Standa

Questions.

a)Identify the persona in the poem . (2mks)

.....
.....
.....

b) What is the message of the poem ? (4mks)

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.....
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.....
.....
.....

(c)Identify and comment on any two aspects of style in the poem . (6mks)

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d) Describe the tone of the poem . (2mks)

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.....
.....
.....

e)Identify and illustrate two character traits of Mwananchi . (4mks)

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.....
.....
.....
.....

(f)Explain the meaning of the following words and phrases as used in the poem . (2mks)

(i) inflated
stomach.....

(ii) Scare you out of your wits
.....

3.GRAMMAR (20mks)

(a)Choose the correct words from brackets to fill in the gaps in the following sentences .

- 1.Wanjiru , together with Omondi and Keter , _____ here every Sunday . (worships , worship)
- 2.Upeo Institute , in conjunction with Lenga Juu Academy and the Ministry of Education , _____ sponsored this event . (has ,have)
3. Mangoes , in addition to oranges , apples and pawpaws _____ served at dinner time .(was , were)
- 4.Nairobi University , as well as Moi University and Kenyatta University , _____ a law degree . (offers ,offer)
- 5.Mary and Stella , including this little boy , _____ to this class.(belongs , belong)

(b)Join the following sentences using the correct relative pronouns .(5mks)

- 1.That is Mrs Wepungulu .She is my maths teacher.
.....
- 2.This disease is difficult to treat .The cause is not known .
.....
- 3.Wangari Maathai has done a lot to conserve the environment .I regard her with respect.
.....
- 4.The egg was rotten .I boiled it.
.....
- 5.Divorce is traumatizing event .It affects the spouses and their children .
.....

(c) Choose the correct form of the adjectives in brackets to fill in the blank spaces (5mks)

- 1.The weather today is _____ than yesterday’s . (forbidding)
- 2.This is the _____ student in this class . (timid)
- 3.I could do with a _____ brush ; this one is worn out . (fine)
- 4.Which is the cake with the _____ sugar ? (little)
- 5.Is this one really as _____ as that one ? (bad)

(d) Use the correct form of the word in brackets to complete each sentence . (5mks)

1. _____ from school without a good reason will be punished .(absent)

2. _____ is what cost him his job . (drunk)

3.The boulders have _____ precariously for years . (hang)

4.They left the scene _____ . (haste)

5.We did it as a matter of _____. (formal)

NAME: ADM NO.: CLASS:

311/1

HISTORY

TERM 1 YEAR 2020

Answer all the questions in the spaces provided.

SECTION A: 20 MARKS

1. Identify the historical period when oral tradition was the main source of information on History and government. (1 mk)
2. State any two characteristics of government. (1 mk)
3. Give two cultural activities that the Agikuyu learnt from the Gumba. (2 mks)
4. What is national integration? (1 mk)
5. Outline any two methods of trade. (2 mks)
6. State two advantages of a bicycle. (2 mks)
7. Outline one feature of independence constitution. (1 mk)
8. Identify the main feature of the system of education during the colonial period in Kenya.(1 mk)

9. Name the Asians who took part in struggle for independence in Kenya. (2m ks)
10. Highlight one factor that undermined the formation of trade unions in Kenya upto 1914. (1 mk)
11. Name the country that was blamed for the outbreak of the 2 world war. (1 mk)
12. Give the immediate cause of the Second World War. (1 mk)
13. State any two features of commonwealth countries. (2 mks)
14. Name the headquarter of African Union. (1 mk)

SECTION B: (30 MARKS)

Answer any two questions from this section.

15. (a) State any three types of Versailles Treaties. (3 mks)
- (b) Explain six reasons why the Allies won the world war I. (12 mks)
16. (a) Outline the demands Austria presented to Serbia following the assassination in Sarajevo. (3 mks)
- (b) What were the achievements of the League of Nations? (12 mks)
17. (a) Give three methods used by Samori Toure during his war of resistance against the French. (3 mks)
- (b) Discuss how judicial independence is protected in Kenya. (12 mks)

TERM 1 2020

c) What is meant by the term apical dominance?

(1mk)

6. For each of the following traits state whether it is continuous or discontinuous (5mks)

Characteristic type of variation

- | | | |
|------|---------------------------------|-------|
| i. | Size of the breast | |
| ii. | Blood groups in man | |
| iii. | Finger prints | |
| iv. | Size of cobs in maize | |
| v. | Ability to taste phenylthiourea | |

Explain the meaning of the following terms in human reproduction (3mks)

- I. Implantation.

- II. Ovulation.

- III. Parturition.

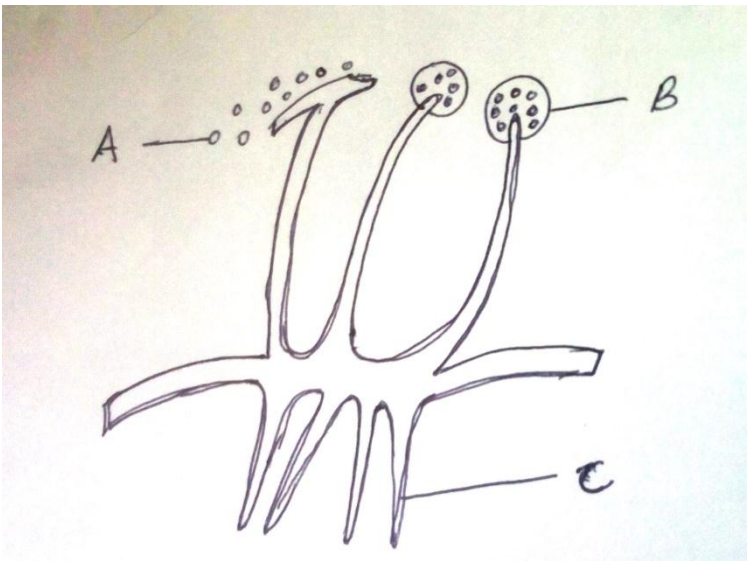
7. In a certain species of plants, the gene for red flowers is codominant to the gene for white flowers. In one experiment of such plants all F1 plants had pink flowers.

a) Using appropriate letter symbols work out the genotypes of the F1 offspring (4mks)

b) If one of the F1 offspring was crossed with a white parent, what would be the phenotypic ratio of their offspring (3mks)

c) what type of cross is shown by the set up in (b) (1mk)

7. The diagram below represents a mature bread mould



Name the structures A, B and C (3Mks)

A.....

B.....

C.....

b)what is the physiological significance of having testes outside the body of the human male?(2mks)

c) What do you understand by the term double fertilization in plants? (2mks)

8). In human haemophilia, is a sex linked ,caused by recessive gene which exerts its effect when in homozygous state.A man whose mother was haemophilic marries a normal woman whose father was haemophilic. **H** represents non haemophilic , **h** represents haemophilia

a)What are the possible genotype of

i. The man.....(1mk)

ii. The woman.....(1mk)

b) Showing your working, find out the possible genotypes of their F1. (4mks)

c)(i) what is the probability that their first born son is haemophilic (1mk)

(ii) what is the probability that their first born child is normal?(1mk)

d) Describe how a breeder can know the genotype of F1 showing dominant phenotype (2mks)

FORM FOUR GEOGRAPHY

TERM ONE, 2020

NAME.....CLASS.....ADM NO.....

1.

- a. Give three economic factors that affect agriculture (3marks)
- b. State three physical factors that favour coffee farming in brazil (3 marks)
- c. Explain four differences between coffee farming Kenya and coffee farming in brazil(8smarks)
- d. Identify the temperate regions where transhumance is practiced. (5marks)

2.

- a. What are choropleth maps? (2marks)
- b. State three disadvantages of using dot maps to present statistical data (3marks)

3.

- a. Differentiate between land reclamation and land rehabilitation (2marks)
- b. Outline four methods used in land rehabilitation (4marks)
- c. State four objectives of reclaiming parts of Yala-Bunyala swamp (4 marks)

4.

- a. Name two rivers that provide irrigation water for Mwea Tebere Irrigation Scheme (2marks)
- b. List three blocks that form mwea irrigation scheme (3 marks)
- c. Explain four physical problems facing irrigation farming in Kenya (3marks)
- d. Explain four benefits of the zuyder zee project in the Netherlands (8marks)

JINA;.....NAMBARI;.....DARASA.....

MJARABU WA KWANZA

MUHULA WA KWANZA,2020

KISWAHILI KIDATO CHA NNE

UFAHAMU

Soma taarifa ifuatayo kisha ujibu maswali.

Mchezo wa kandanda umenifunza jambo moja: mashabiki ndio wachezaji bora sana. Wao hawakosei kufunga bao au “kuhata” penalti wanavyosema magwiji wa mchezo huu wenyewe. Mchezaji akiukosea mpira, mashabiki hunung’unika na kupiga yowe. Katika hilo yowe, huwa kumebewa makombora mengi. Yapo ya matusi, yapo ya “mwondoeni nje ya uwanja” yapo ya “kocha lazima aende,” na yapo ya “asipewe nambari tena” wajua tena kandanda!

Tunaweza kutumia kandanda kujifunza mengi mengine maishani. Kwa mfano, juu ya sheria na haki za watu. Hebu fikiria, mashabiki kufanya yowe hata timu ikashinda kombe! Au hata kufanya yowe hata mchezaji akabadilishwa na wengine bora wakaletwa uwanjani? Isitoshe, shabiki ambaye ni mfalme wa uwanja, yowe lake linaweza hata kumfanya kocha wa kilabu maarufu duniani kama Manchester United au Arsenal au Real Madrid kuondoka! Mbona hatufanyi hivyo dhidi ya sheria zinazovuruga maisha ya ndugu zetu?

Mbona hatupigi yowe dhidi ya ubakaji? Wapi yowe mwanamke anapopigwa na mumewe hadi kufa? Wapi yowe mtoto wa kike anaponyimwa urithi kwa sababu ya sheria potovu za kitamaduni? Wapi yowe dhidi ya wizi wa kimabavu?

Watoto wetu leo hawawezi kutembea peke yao haswa magharibi yanapoingia. Je, hatuwajui watu wanapowahatarisha wana wetu? Tunafanya kitu gani? Mbona hatupigi yowe polisi au utawala ukasikia ili hatua ichukuliwe?

Si kitambo tuliposikia kuhusu biashara ya ngozi ya binadamu kuuzwa ghali sana, hasa katika nchi jirani ili waganga watengenezee dawa za kuwapumbazia waja. Huu ni uchawi na kilele cha ushetani. Katika nchi jirani juzijuzi, walishikwa watu wawili ambao walikuwa wamechinja mtoto wa kiume na kumchuma ngozi kwa ushirikiano na mpumbao mkubwa. Yaani tumeuza za wanyama zikatushinda, sasa tunauza za watu?

Tatizo kubwa ni kwamba watu wanaofanya biashara haramu tunawajua. Je, tunachukua hatua gani? Haijalishi wanavaa nguo gani, za fisi au za mwanakondoo! Tunahitaji kuwaweka wazi ili biashara zao zijulikane. Wawe weusi au manjano, weupe au kijani, sharti weupe uwepo kuondoa hofu na taharuki kwa ajili ya kila mwanajamii.

Hatua ya kwanza itakuwa ya kugeuza sera na mtazamo wa vikosi vyetu vya usalama kuhusu upokeaji wa habari kutoka kwa umma. Endapo hawatasita kufichua wanaowapa habari, watajikuta katika hatari ya kuonwa na raia kama wapinga amani na lazima wachukue lawama kwa uongezekaji wa uhalifu nchini. Kwa wazalendo hata hivyo, yowe ni silaha yetu.

Maswali

- a) Yape makala haya kichwa mwafaka. (alama 1)

.....

.....

- b) Mchezo wa kandanda una mchango gani? (alama 2)

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.....

- c) Ni matatizo gani ambayo mwandishi anailaumu jamii kwayo? (alama 3)

[illegible]

- d) Lawama kubwa inaegemea kina nani? Taja mawili. (alama 2)

- eTaja mbinu zozote mbili za lugha zilizotumiwa na mwandishi. (alama 2)

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MATUMIZI YA LUGHA

- (a) Tumia neno Kamene kama yambiwa katika sentensi. (alama 2)

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.....

- (b) Taja misingi ya upangaji wa irabi za Kiswahili. (alama 3)

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.....

(c) Huku ukitoa mfano mmoja, eleza maana ya mofimu ya kimsamiati. (alama 2)

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(d) Vitambue vijenzi vya kimsingi vya lugha ya Kiswahili. (alama 2)

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.....

(e) (i) Eleza maana ya kishazi. (alama 1)

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.....

(ii) Tunga sentensi moja yenye kishazi tegemezi kinachovumisha. (alama 2)

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.....

(f) Badilisha kitenzi kisaidizi kilicho kwenye sentensi ifuatayo kiwe kishirikishi.
(alama 2)

Mvule alikuwa akilala bwenini.

.....
.....

(g) Taja majukumu ya vipengele vilivyopigwa kijistari. (alama 4)

Nanyi nendeni haraka au muadhibiwe kinyama kwa utundu wenu.

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(h) Andika sentensi ifuatayo katika ukubwa. (alama 2)

Uteo ulionunuliwa umezeeka sana.

.....

I Tunga sentensi moja mwafaka kudhihirisha matumizi ya kiunganishi madhali.

(alama 2)

.....

.....

(j) Onyesha minyambuliko ya maneno yafuatayo katika kauli zilizotolewa mabanoni.
(alama 2)

(i) ja (tendewa)

(ii) cha (tendeka)

(k) Katika sentensi moja, onyesha matumizi mawili ya kibainishi. (alama 2)

.....

.....

.....

(l) Yakinishi (alama 2)

Usipolipa usiingie ndani

.....

.....

(m) Neno 'gofu' linapatikana katika ngeli gani? (alama 1)

.....

(n) Geuza sentensi ifuatayo iwe katika usemi halisi. (alama 2)

Kassim alimwomba Hirshi aende mahali alipokuwa ili amtume.

.....

.....

(o) Changanua sentensi ifuatayo kwa kutumia kielelezo matawi. (alama 4)

Japo Mutiso anasinzia darasani hajamaliza kazi yake.

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ISIMUJAMII (Alama 5)

Eleza sifa tano za kimsingi zinazotambulisha sajili ya siasa. (alama 5)

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**2020 EXAMINATION
FORM 4 TERM 1 2020**

NAME:**ADM:**.....**DATE:**.....

1. Use logarithms to 4 decimal places to evaluate: (4 marks)

$$\left(\frac{0.7841 \times \sqrt{0.1356}}{\text{Log } 84.92} \right)^{\frac{1}{3}}$$

2. A globe representing the earth has a radius of 0.5m. point A(0°, 10°W), B (0°, 35°E), P(60°N, 110°E) and Q(60°N, 120°W) are marked on the globe.

- a) Find the length of arc AB, leaving your answer in term of π (3mks)

3. A circle centre is the point C(2,3) passes through a point P(a,b). A point M(-2, $-\frac{5}{2}$) is the mid-point of the line CP.

- a) Calculate the coordinates of P. (1mk)

- b) Determine the equation of the circle in the form $x^2 + y^2 + ax + by + c = 0$ (3mks)

4. Make **a** the subject of the formula:

$$x = y + \sqrt{x^2 + a^2} \quad (3\text{marks})$$

5. Given that $\sin\left(\frac{2}{3}x + 20^\circ\right) - \cos\left(\frac{5}{6}x + 10^\circ\right) = 0$. Without using a mathematical table or a calculator, determine $\tan(x + 20^\circ)$. (3 marks)

6. Two fair dice one a regular tetrahedron (4 faces) and the other a cube are thrown. The scores are added together. Complete the table below to show all possible outcomes. (2mark)

		CUBE					
		1	2	3	4	5	6
TETRAHEDRON	1						
	2						
	3						
	4						

a) Find the probability that:

i) The sum is 6. (1 mark)

iii) The sum is 6 or 9. (2 marks)

7. A particle moves along a straight line such that its displacement s metres from a given point is $s = t^3 - 5t^2 + 3t + 4$ where t is time in seconds. Find:
- (a) The displacement of the particle at $t = 8$. (2 marks)
- (b) The velocity of the particle when $t = 10$. (3 marks)
8. A classroom measures $(x + 2)$ m by $(x - 5)$ m. If the area of the classroom is 60m^2 . Find its length. (3 marks)

SECTION B

Lengths of 100 mango leaves from a certain mango tree were measured to the nearest centimeter and recorded as per the table below,

Length in cm	No. of leaves
10 to 12	3
13 to 15	16
16 to 18	36
19 to 21	31
22 to 24	14

- a) On the grid provided draw a cumulative frequency graph to represent this data. (5mks)
- b) Use your graph to estimate
- i) The median length of the leaves (2mks)
- ii) The number of leaves whose lengths lie between 13cm and 17cm. (3mks)

- (a) Draw $\triangle PQR$ whose vertices are $P(1, 1)$, $Q(-3, 2)$ and $R(0, 3)$ on the grid provided.(1mk)

- (b) Find and draw the image of ΔPQR under the transformation whose matrix is $\begin{pmatrix} 3 & 0 \\ 1 & 1 \end{pmatrix}$ and label the image $P^1Q^1R^1$. (2 marks)
- (c) $P^1Q^1R^1$ is then transformed into $P^{11}Q^{11}R^{11}$ by the transformation with the matrix $\begin{pmatrix} -1 & 0 \\ 1 & 3 \end{pmatrix}$. Find the co-ordinates of $P^{11}Q^{11}R^{11}$ and draw $P^{11}Q^{11}R^{11}$. (3 marks)
- (d) Describe fully the single transformation which maps PQR onto $P^{11}Q^{11}R^{11}$ find the matrix of this transformation. (3 marks)

- (d) Describe fully the single transformation which maps PQR onto $P^{11}Q^{11}R^{11}$ find the matrix of this transformation. (3 marks)

NAME..... INDEX NO.....

STUDENT'S SIGN..... DATE.....

232/1
PHYSICS
FORM 4
2020

TIME: 2 HOURS

Physics
Paper 1

INSTRUCTIONS TO THE CANDIDATES:

- Write your **name and admission number** in the spaces provided above.
- Answer **all** the questions both in section **A** and B in the spaces provided below each question
- All workings **must** be clearly shown; marks may be awarded for correct steps even if the answers are wrong.
- Mathematical tables and silent electronic calculators may be used.
- Take gravitational acceleration = 10m/s^2 and $71=3.142$.

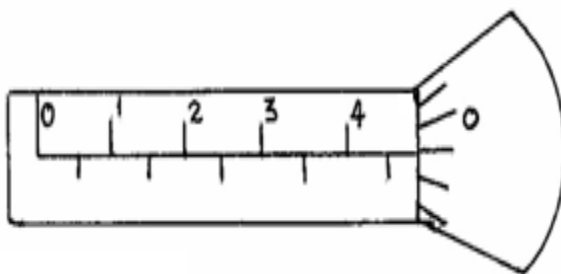
FOR EXAMINER'S USE ONLY

SECTION	QUESTION	MAXIMUM SCORE	STUDENT'S SCORE
A	1 – 11	25	
B	12	11	
	13	12	
	14	10	
	15	13	
	16	09	
TOTAL		80	

SECTION A (25 MARKS)

Answer all questions in this section in the spaces provided:

1. The diagram **below** shows an instrument used by a student to measure the thickness of a wire.



- (a) Name the instrument. (1 mark)

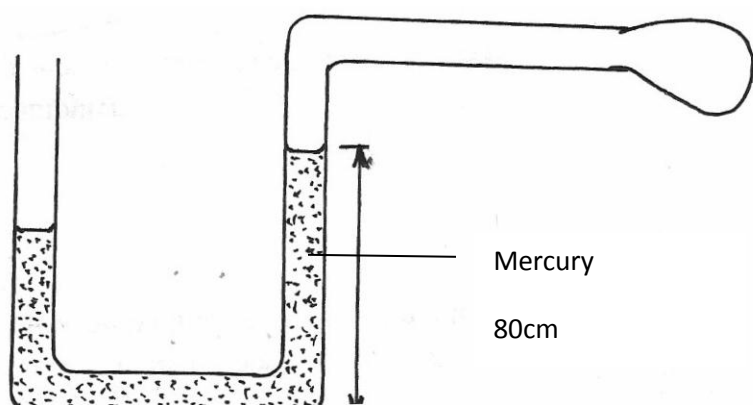
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- (b) If it has a zero error of 0.08mm, what is the actual thickness of the wire? (2marks)

2. A radio signal of 30MHz is received by an aerial whose length is $\frac{1}{8}^{\text{th}}$ of its wavelength. If the speed of light is 3.0×10^8 m/s determine the length of the aerial. (3 marks)

3. A drop of olive oil, whose volume is 0.12 mm^3 , was placed on a surface of clean water. The oil spread and formed a patch of area $6.0 \times 10^4 \text{ mm}^2$. Estimate the size of the olive oil. (4 marks)

4. The figure below shows the apparatus used to examine the pressure of a gas.

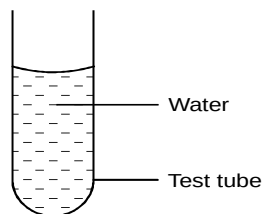


Taking density of mercury to be $13,600\text{kg/m}^3$ standard atmospheric pressure $100,000\text{N/m}^2$ (3marks)

5. Explain why a horse rider bends forward when the horse is on high speed. (1 mark)

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6. Explain why water curves in a test as shown in the figure below. (1 mark)



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7. A car is brought to rest from 180km/h in 25 seconds. What is its retardation? (3 marks)

8. (a) State Newton's Law of Momentum.

(1 mark)

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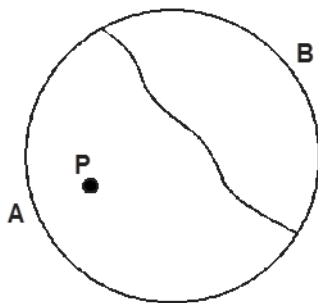
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(b) Give the SI unit for momentum.

(1 mark)

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9. The figure below shows a thin thread tied on the surface of water in trough.



When small drops of soapy water was dropped at point P, the thread curved towards side B. Explain?

(2 marks)

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10. List two factors that affect the melting point of ice

(2 marks)

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11. Explain why water spilled on a glass surface wets the surface.

(1mark)

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SECTION B (55 MARKS)

Answer all questions in this section.

12. (a) Define "trajectory".

(1 mark)

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(b) A ball is thrown from the top of a cliff 20m high with a horizontal velocity of 10m/s. Calculate,

(i) The time taken by the ball to strike the ground. (2 marks)

(i) The distance from the foot of the cliff to where the ball strikes the ground. (2 marks)

(ii) The vertical velocity at the time it strikes the ground. (take $g=10\text{m/s}^2$) (1 mark)

(c) A ball of mass 100g is dropped from a height of 1.25m above the ground surface. It rebounds to a height of 1.1m. Calculate

(i) Velocity of ball before impact (2 Marks)

(ii) Force of impact (Take $g = 10\text{N/kg}$) (3 Marks)

13. a). State two differences between heat transfer by convection and radiation (2 marks)

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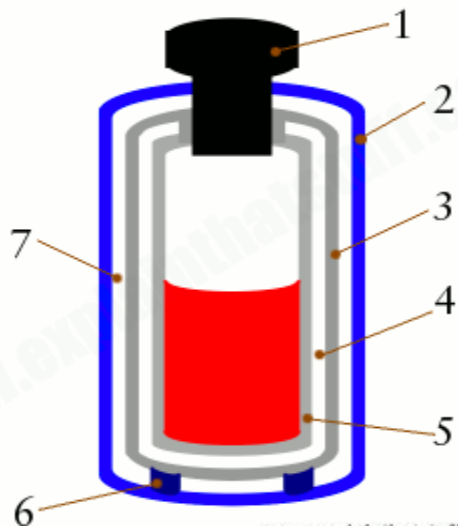
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b). Give a reason why a thick glass bottle cracks when boiling hot water is suddenly poured inside it (1mark)

(c) State **two** applications of convection.

(2 marks)

d) Below is a vacuum flask. Use it to answer questions that follow.



(i) Name the parts labeled.

(3 marks)

- I. 4
- II. 1
- III. 2

(ii) Give the function of the parts labeled:

(4 marks)

- I. 6
- II. 5
- III. 7
- IV. 3

14. (a) A uniform rod of length 1.0 m is hung from a spring balance and balanced in horizontal position by a force of 1.6 N. Determine;

i) The weight of the rod.

(3 marks)

(ii) Reading of the spring balance. (2 marks)

b)(i) What is equilibrium? (1 mark)

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(ii) State two conditions that must be satisfied for a body to be in equilibrium. (2 marks)

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(c) Mention **two** factors that affect stability of an object. (2 marks)

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15. (a) An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm.

(i) Draw the ray diagram of the image formed. (2 marks)

(ii) Determine the position of the image formed. (3 marks)

(iii) Determine size of the image formed. (3 marks)

(iv) With a reason, what is the nature of the image formed? (2 marks)

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(b)(i) Identify the type of curved mirror used as a driving mirror. (1 mark)

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(ii) Give **two** properties of the type of mirror in b(i) above that make it suitable for that purpose. (2 marks)

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16. (a) State the relationship between electricity and magnet. (1 mark)

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(b) Below is procedure of Oersted's experiment. Use it to answer questions that follow.



Procedure

1. Cut a 1 meter loop of insulated wire.
 2. Use electrical tape to secure a stripped end of the wire to one side of a D battery.
 3. Run the wire up one side of the box, across the top, and down the other side. Make sure you have enough wire so that it can run along the table or ground to reconnect the battery. Now you have a loop!
 4. Connect the other open end of the wire to the battery so current begins to flow.
 5. Bring the compass into the center of the loop. What happens?
 6. Move the compass around closer to the wire and away from the wire. Record your observations.
- (i) What will happen when you bring the compass towards the current loop? (1 mark)
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- (ii) Explain (b) (i) above. (1 mark)
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- c) Mention two ways in which an electromagnet can be made stronger. (2 marks)
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- (b) Give two factors affecting force on a current-carrying conductor in a magnetic field. (2 marks)
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- (c) Give two areas where knowledge of electromagnetism can be applied. (2 marks)

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FOR ANSWERS AND MORE SETS OF EXAMS, SMS OR WHATSAPP:

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